Software Requirements Specification

for

Hostel Management and Complaint Registration

Version 1.0 approved

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Revision History

Name	Date	Reason For Changes	Version

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1. Introduction

1.1 Purpose

The purpose of creating this SRS is to provide a clear and comprehensive understanding of the project for all stakeholders involved, including developers, designers, testers, and clients. This document describes the project's target audience and its user interface, hardware and software requirements. It defines how students, hostel staff and our team see the product and its functionality.

1.2 Document Conventions

- The body text shall be justified and typed in the Font Style 'Times New Roman' and Font size 12.
- Heading shall be typed in the Font Style 'Times New Roman', Font size 18 and bold.
- Subheading shall be typed in the Font Style 'Times New Roman', Font size 14 and bold.
- Bulleted list shall be used whenever order of points is not important.
- Numbered list shall be used whenever order of points is important.

1.3 Intended Audience and Reading Suggestions

This Software Requirements Specification is primarily intended for the developers, students, hostel management staff, testers and other concerned parties. This requirements document is organized into several sections that can be read and referenced as needed.

Sections to be read for developers:

1. Introduction- 1.1 Purpose, 1.4 Product Scope
2. Overall Description- 2.1 Product Perspective, 2.2 Product Functions, 2.3 User Classes and Characteristics
3. External Interface Requirements- 3.1 User Interfaces, 3.3 Software Interfaces, 3.4 Communication Interfaces
4. System Features- All
5. Other Non-Functional Requirements- All

Sections to be read for users:

1. Introduction- 1.3 Intended Audience and Reading Suggestions, 1.4 Product Scope
2. Overall Description- 2.4 Operating Environment, 2.5 Design and Implementation Constraints, 2.6 User Documentation, 2.7 Assumptions and Dependencies
3. External Interface Requirements- 3.1 User Interfaces, 3.2 Hardware Interfaces
4. System Features- None
5. Other Non-Functional Requirements- 5.1 Performance Requirements

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1.4 Product Scope

This Hostel Management and Complaint Registration website aims to revolutionize hostel administration by leveraging technology to improve efficiency, communication, and overall student satisfaction.

Purpose:

- The primary purpose of this software is to provide an integrated platform for efficient hostel administration and student complaint handling**.**

Benefits:

- Increased Operational Efficiency: Automation of routine tasks leads to a more efficient hostel management process.
- Enhanced Student Satisfaction: Prompt complaint resolution and improved communication contribute to a positive student experience.

Objectives:

- Optimize Hostel Operations: Streamline hostel management processes to ensure efficient room allocation, student and hostel data management, and resource utilization.
- Facilitate Complaint Registration: Provide students with a user-friendly platform to register complaints, feedback, and requests effortlessly.
- Ensure Data Security and Compliance: Implement robust security measures to protect student information and ensure compliance with data protection regulations.

Goals:

- User Engagement: Achieve a high level of user engagement by creating an interface that is intuitive, responsive, and user-friendly.
- Scalability: Design the system to accommodate future growth in the number of students and additional features, ensuring long-term viability.
- Data Accuracy: Ensure the accuracy and integrity of resident data by implementing validation checks and data verification processes.
- Accessibility**:** Develop the software with accessibility features to cater to a diverse user base, including those with different abilities.

By aligning with corporate goals or business strategies, the Hostel Management and Complaint Registration Software contribute to the efficient and effective management of hostel facilities, ultimately improving the overall living experience for residents and optimizing administrative processes.

1.5 References

https://www.slideshare.net/hiraakram11/hostel-management-system-srs
https://studylib.net/doc/25437171/355718494-151838501-hostel-management-srs-doc-1-pdf

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2. Overall Description

2.1 Product Perspective

Our Hostel Management and Complaint Registration system is a fully independent software system. It is not a part of any other system.

2.2 Product Functions

The product should be able to perform following functions:

- User Authentication and Authorization:
  - Secure login for administrators, staff and students.
  - Role-based access control to manage permissions.

- Hostel Management:
  - Room allocation and deallocation functionality.
  - Room occupancy tracking and reporting.
  - Listing contact information of hostel staff.

- Complaint Registration and Management:
  - User-friendly complaint registration interface for students.
  - Categorization of complaints based on type of maintenance needed.
  - Ability to track complaint resolution process (registered, in-process, resolved)

- User Information Management:
  - Administrators should be able to add, edit or delete student information.
  - Hostel staff should be able to use this student information for allocation of rooms and to keep track of the students.

- Room and Facility Information:
  - Administrators should be able to add, edit or delete rooms of a particular hostel.
  - Hostel staff should be able to provide a list of facilities to administrators to display on the site.
  - Display information about available rooms, hostel facilities, rules and regulations.

- Data Security and Privacy:
  - Encryption of sensitive data.
  - Compliance with data protection regulations.

- Help and Support:
  - Support system for issue resolution.

- Logout and Session Management:

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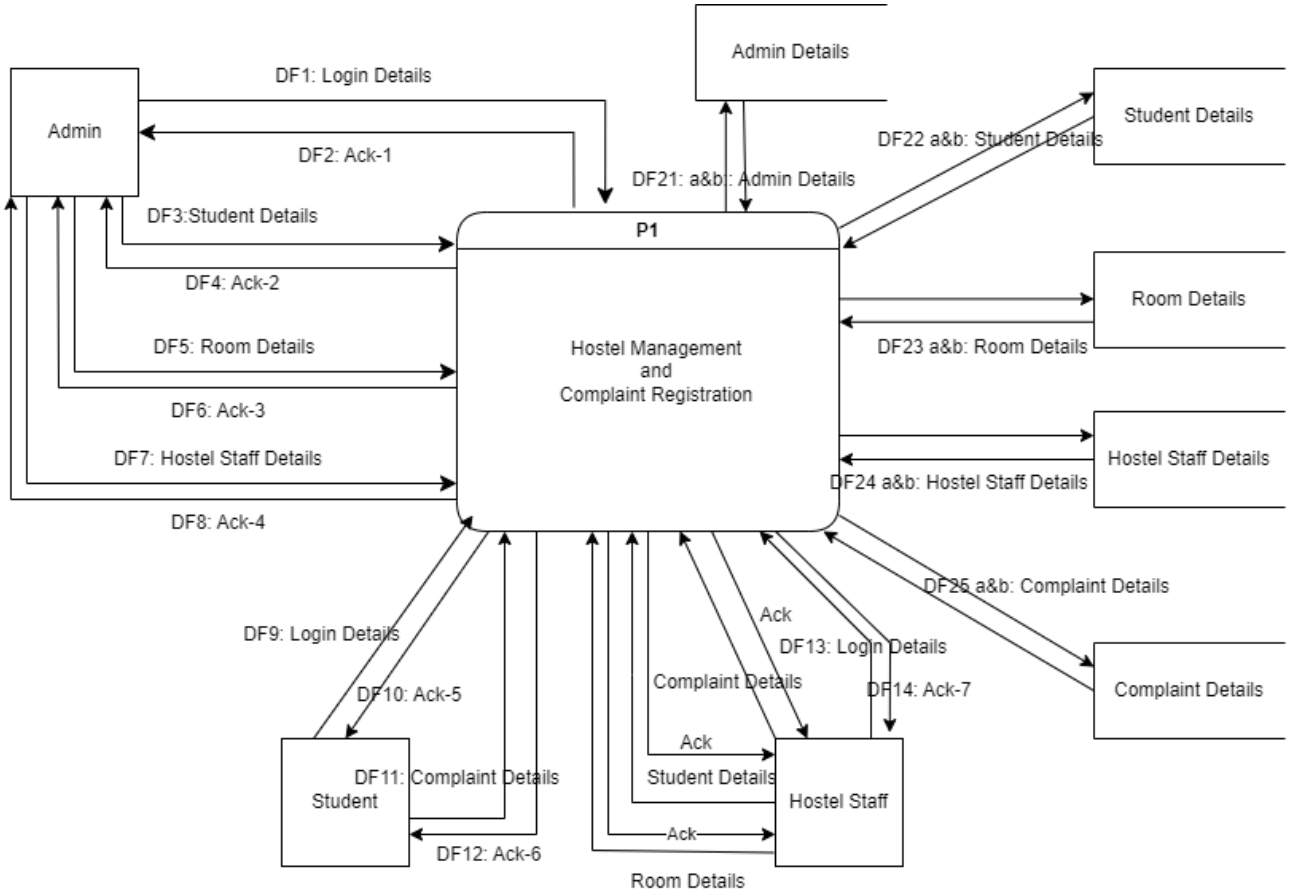
- Secure logout functionality.
- Session timeout and management.



2.3 User Classes and Characteristics

There are three kinds of users for the proposed system:

1. Administrator
2. Hostel Staff/Manager
3. Student

Characteristics of Administrator:

- Administrators have access to all system functionalities and data. This includes the ability to configure, manage, and monitor the entire system.
- They are responsible for managing user accounts, including creating new accounts, modifying permissions, and deactivating accounts when necessary.
- Assigns roles and permissions to other users based on their responsibilities within the system. This includes assigning roles to staff, students, and other administrators.

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- Undertakes routine maintenance tasks, such as updates, to ensure the system's reliability and security.
- Manages the overall data within the system, including overseeing data storage, data security, and data integrity.
- Addresses issues related to system functionality and user access promptly. They are the first point of contact for addressing any technical problems.
- Provides training to other administrators and users as needed. Offers support to users who may face challenges in using the system effectively.
- Monitors and enforces security measures within the system. This includes implementing and overseeing access controls, authentication, and authorization processes.

Characteristics of Hostel Staff/Manager:

- Handles the assignment of rooms to students.
- Manages and oversees day-to-day hostel operations, including room allocation, check-ins, and check-outs.
- Responsible for ensuring that the hostel facilities are well-maintained and meet the standards set by the institution.
- Monitors and manages room occupancy, ensuring that rooms are allocated efficiently and that any vacancies are addressed promptly.
- Provides guidance and support to students regarding hostel policies, procedures, and system usage. Offers assistance to students who may have questions or issues.
- Collaborates with the Complaint Resolution Team to address and resolve escalated complaints effectively.
- Collaborates with administrators to ensure that the hostel management system aligns with overall institutional goals and policies.

Characteristics of Students:

- Can submit room preferences through the system during specific periods.
- Students have the ability to register complaints related to hostel facilities and services through the system.
- Has access to personal information within the system, including room details, complaint history, and contact information.
- Accesses information about hostel facilities, rules, and regulations.
- Can check the status and progress of registered complaints and requests through the system.

2.4 Operating Environment

Hardware Platform:

- No specific hardware requirements for users' personal devices.
- Better to use the website on PC or Desktop.

Operating System:

- Compatible with Windows, macOS, and Linux on users' personal devices.

Web Browser:

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- Compatible with modern web browsers such as Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari.

Internet Connection:
- Users should have a stable internet connection for optimal performance.

Software Dependencies:
- No additional software dependencies on users' devices; the website runs on standard web technologies.

Database:
- Our system shall be using MongoDB for storing all the data.

Hosting Platform:
- Our front-end and back-end both shall be hosted on Vercel.

2.5 Design and Implementation Constraints

- The development of the system will be constrained by the availability of required software such as web servers, database and other development tools.
- The hardware constraints include a PC or desktop is required to access the website and use its services.

2.6 User Documentation

We shall post videos regarding how to utilize the software for students and hostel staff. Also help will be provided within 2 days of contacting us.

2.7 Assumptions and Dependencies

- We have assumed three types of users of our system as discussed above.
- Details related to hostels, students, hostel staff.
- Roles and Tasks are predefined.
- Basic computer skills are required by the user to use the website.

3. External Interface Requirements

3.1 User Interfaces

1. General Interface

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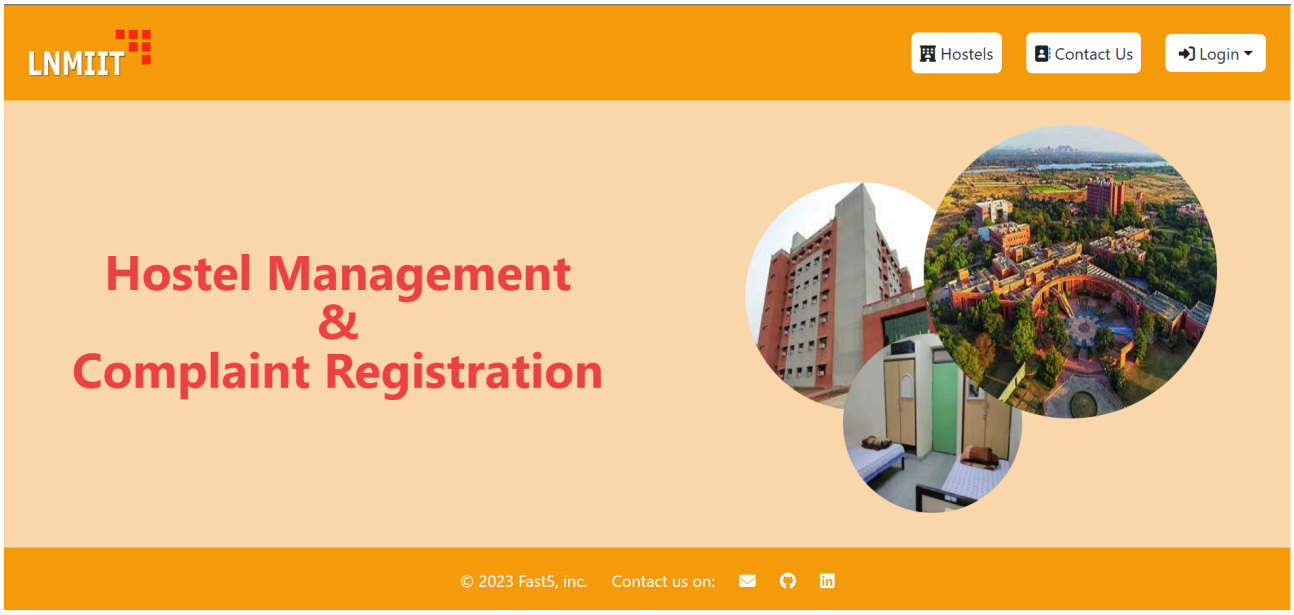
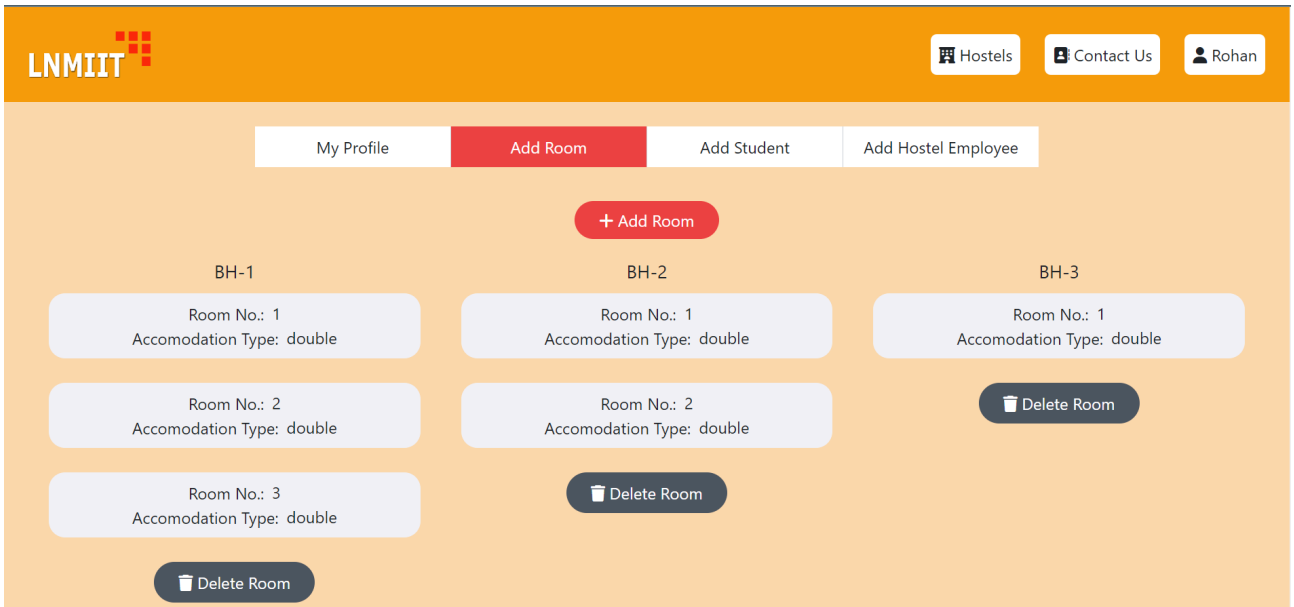


2. Admin Interface



3. Hostel Staff Interface
4. Student Interface

GUI standards:

- Consistency: Maintain a consistent layout throughout the application.
- Navigation: Intuitive navigation with standardized menu structures.
- Error Handling: Clear and concise error messages.

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3.2 Hardware Interfaces

1. General Interface:
   Logical Characteristics:
   - Provides a unified, simplified view for all users.
   - Access to general hostel information, and contact us.

   Physical Characteristics:
   - Supported device types: Desktops, and laptops.
   - Communicates with the database, web server, and network using standard web protocols (HTTP/HTTPS).

2. Admin Interface:
   Logical Characteristics:
   - Offers advanced functionalities for administrative tasks.
   - Complete user and hostel management capabilities.

   Physical Characteristics:
   - Supported device types: Desktops and laptops.
   - Direct interaction with the database and server for real-time data control and updates.

3. Hostel Staff Interface:
   Logical Characteristics:
   - Designed for hostel staff members responsible for daily operations.
   - Access to room allocation, hostel rules, and complaint handling features.

   Physical Characteristics:
   - Supported device types: Desktops and laptops.
   - Interaction with the web server and database for data retrieval and management.

4. Student Interface:
   Logical Characteristics:
   - Tailored for students residing in the hostel.
   - Access to complaint registration, personal profile, and other services.

   Physical Characteristics:
   - Supported device types: Desktops and laptops.
   - Communication with the web server through secure web protocols (HTTPS) for data updates and notifications.

3.3 Software Interfaces

Database:
- Database Management System: MongoDB 7.0
- Data Items: Date items include admin information, room information, hostel staff information, student information, complaint records.
- Purpose: The system stores, retrieves, and manages data related to hostel management and complaints using the database.

Operating Systems:

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- Supported OS Versions: Windows 8 and above, Ubuntu 20.04, Android 7 and above
- Purpose: The software product needs to run on these operating systems to serve a broad user base.

Hosting Platform:
- Hosting Service: Vercel Hobby Plan (free plan)
- Data Interactions: Vercel hosts the website and handles incoming HTTP requests from clients.
- Purpose: Vercel ensures the website's availability and manages server infrastructure.

Web Server:
- Web Server Software: Vercel serves as the web server, managing HTTP requests.
- Data Interactions: Incoming HTTP requests from clients, including user login requests and complaint submissions.
- Purpose: Vercel handles client requests, routes them to the appropriate components within the website, and returns responses.

APIs:
- External APIs: Mailchimp
- Data Items: email message.
- Purpose: To send email for notification purposes.

Programming Languages and Frameworks:
- Programming Languages: HTML, CSS, JavaScript for frontend and Node.js for backend.
- Libraries and Frameworks: React (JavaScript Library), Tailwind CSS and Bootstrap (CSS frameworks) for frontend and Express.js (Node.js framework) for backend.
- Libraries and frameworks facilitate various operations, including user interface design, data processing, and routing.
- Purpose: These components aid in the development and functionality of the website.

Data Sharing:
- Data Sharing Mechanism: RESTful API
- Shared Data: Date items such as Room information, Student information is shared between components.
- Purpose: Data sharing is essential for communication and consistency between website components.

3.4 Communications Interfaces

Web Browser Communication:
- Requirement: Users should interact with the system through a web browser interface.
- Communication Standards: HTTP/HTTPS for web page requests and responses.

Network Server Communications Protocol:
- Requirement: The system should communicate with external services or databases for data retrieval and integration.

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- Message Formatting: Use JSON or other standard data interchange formats.
- Communication Standards: HTTP/HTTPS for RESTful API endpoints.

Electronic Forms:

- Requirement: Users should be able to submit various electronic forms, including registration forms and complaint submission forms.
- Message Formatting: Use HTML forms for user input.
- Communication Standards: HTTP/HTTPS for form submission.

4. System Features

4.1 Add, Edit or Delete Room

4.1.1 Description and Priority

Description: This feature allows the admin to create, edit or delete rooms for each hostel corresponding to the request of the hostel manager.

Priority: High (initially since when there are no rooms) and then priority gradually falls to low.

4.1.2 Stimulus/Response Sequences

Stimulus: Admin selects the option to create a new room.
Response: Website then renders a form page to add details of the new room.

Stimulus: After filling the details, the admin clicks the button to add the room.
Response: Website adds the room and renders a page which displays all the rooms of respective hostels.

Stimulus: Admin clicks the delete button available at different columns of the screen representing different hostels and each column's rows representing rooms of that particular hostel.
Response: The website deletes the last room which was created for that particular hostel and again renders the same page showing the updated list of rooms.

Stimulus: Admin clicks on any room of any hostel.
Response: Website will show the details of that particular room and will also allow the admin to edit the room (room type – single or double).

Stimulus: After changing the details, the admin clicks the edit room button.
Response: Website then again renders the updated room list.

4.1.3 Functional Requirements

REQ-1: authentication of admin.

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REQ-2: a room form must be filled in order to add room.
REQ-3: rooms must be present in order to edit or delete rooms.
REQ-4: if any error occurs, a popup shall be displayed showing the error message.

4.2 Add, Edit or Delete Student

4.2.1 Description and Priority

Description: This feature allows the admin to create, edit or delete students (when students are admitted).
Priority: High (annually when new sets of students are admitted) and then priority gradually falls to low.

4.2.2 Stimulus/Response Sequences

Stimulus: Admin selects the option to add a new student.
Response: Website then renders a form page to add details of the new student.

Stimulus: Admin then clicks the add new button.
Response: Website will show errors if any like few details missing or student already exists (based on username or roll no). Otherwise, the website renders the page listing all the created students.

Stimulus: Admin clicks on any listed student.
Response: Website will show the details of the selected student and allow the admin to edit any detail of the student or delete the student.

Stimulus: Admin clicks on edit/delete student button.
Response: In both the cases, the website will again render the page showing the updated list of students.

4.2.3 Functional Requirements

REQ-1: authentication of admin.
REQ-2: a student form must be filled in order to add student.
REQ-3: students must be present in order to edit or delete students.
REQ-4: if any error occurs, a popup shall be displayed showing the error message.

4.3 Room Allocation and Deallocation

4.3.1 Description and Priority

Description: This feature allows the hostel management staff to allocate and deallocate rooms to students.

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Priority: High (annually when new sets of students are admitted) and then priority gradually falls to medium.

4.3.2 Stimulus/Response Sequences

Stimulus: Hostel Manager sees a list of un-allocated rooms and then he/she clicks on add student1 and add student2 (in case of a double room) or add student (in case of a single room).
Responses: It would display a list of students who are not allocated any room.

Stimulus: On clicking a student from the unallocated student list.
Responses: Website will again render the updated list of rooms and also the information of that particular student will be updated (room no will be added).

Stimulus: Hostel Manager sees a list of allocated rooms and then he/she clicks on that room to remove a student from that room.
Responses: Website will again render the updated list of rooms and also the information of that particular student will be updated (room no will be removed).

4.3.3 Functional Requirements

REQ-1: authentication of hostel staff/management.
REQ-2: vacant rooms and students must be there in order to allocate or deallocate rooms.
REQ-3: if any error occurs, a popup shall be displayed showing the error message.

4.4 Complaint Registration

4.4.1 Description and Priority

Description: This feature enables the students to raise complaints and allows the hostel management staff to resolve them.
Priority: Medium

4.4.2 Stimulus/Response Sequences

Stimulus: A student clicks the register complaint button.
Response: A form is displayed to the student asking relevant details about the complaint (type, location and so on).

Stimulus: The student then fills the complaint registration form and clicks the button to register a complaint.
Response: Then the website renders the page showing the updated list of complaints registered by that student (resolved, unresolved and pending).

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Stimulus: The complaint made by that student then appears in the unresolved list of complaints of the hostel management account. Then, the hostel manager can mark it as pending (status).
Response: Website then again renders the page showing the updated list of unresolved complaints and also for the student, the complaint is shown as marked as pending.

4.4.3 Functional Requirements

REQ-1: authentication of students.
REQ-2: a complaint registration form must be filled in order to register a complaint.
REQ-3: a complaint must be marked as registered to mark it pending or resolved.
REQ-4: if any error occurs, a popup shall be displayed showing the error message.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

Page Load Time:

- Requirement: The website should load within 3 seconds for 90% of users under normal traffic conditions.
- Rationale: Fast page load times improve user experience and reduce bounce rates. Users expect quick access to hostel information and complaint registration.

Concurrent User Support:

- Requirement: The system should support a minimum of 100 concurrent users without performance degradation.
- Rationale: Ensuring the website can handle concurrent users is essential for providing services to a hostel with a large number of residents and staff.

Complaint Registration Response Time:

- Requirement: When registering a complaint, the system shall provide a response within 2 seconds to acknowledge the submission.
- Rationale: Quick acknowledgment assures users that their complaints are received, increasing user satisfaction.

Error Handling:

- Requirement: Error messages should be delivered within 2 seconds of an error occurrence, with clear details and suggested actions.
- Rationale: Swift error handling helps users resolve issues and reduces frustration.

5.2 Safety Requirements

- Passwords of all the types of users shall be encrypted before storing in the database.

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- Information from the database can only be accessed by the authorized user.

5.3 Security Requirements

User Authentication and Authorization:

- Define user roles and their access rights (e.g., admin, staff, students).
- Define role-based access control (RBAC) to manage permissions effectively.
- Implement secure authentication mechanisms to ensure only authorized users can access the system.
- Ensure password policies (e.g., complexity requirements that password must contain 8 characters) are in place.

Session Management:

- Implement secure session management to prevent session hijacking or fixation.
- Define session timeout settings to automatically log users out after a period of inactivity.

Input Validation:

- Enforce strict input validation to prevent SQL injection, Cross-Site Scripting (XSS), and other injection attacks.

5.4 Software Quality Attributes

Performance: The software should be developed such that the performance or average turn-around time of response after requesting a service should remain high even in the worst-case scenarios.

Usability: The software should be designed and developed keeping in mind its user base.

Testability: The codebase shall be designed with modular components to facilitate unit and integration testing.

Maintainability: The software system should be developed in such a manner such that it takes less person-days of effort to maintain (in case of error or adding or improving functionality) the software after delivery.

Availability: All the services of the system should be available to all its intended users at all times.

Portability: The software system should remain consistent with all kinds of web browsers.

5.5 Business Rules

User Roles and Permissions:

- Users are categorized into different roles (e.g., general, administrators, hostel staff, students).
- Each role has specific permissions and access rights (e.g., administrators can manage users, while students can only submit complaints).

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Complaint Submission:

- Only registered students can submit complaints.
- Complaints must include relevant information (e.g., description, location, category).
- Complaints should be time stamped upon submission.

Complaint Management:

- Hostel staff can view and manage complaints.
- Complaints shall be assigned to maintenance staff for resolution.
- Complaint status (open, in progress, resolved) should be updated as necessary.

Room Allocation:

- Room allocation should follow specific rules and criteria (e.g., preference, room type, availability).
- Students may request room changes under certain conditions (e.g., availability, approval by hostel management staff).

6. Other Requirements

Appendix A: Glossary

HMCRS: Hostel Management and Complaint Registration System
SRS: Software Requirement Specification
HTML: Hyper Text Markup Language
CSS: Cascading Style Sheet
HTTP: Hypertext Transfer Protocol

Appendix B: Analysis Models

Level-0 DFD:

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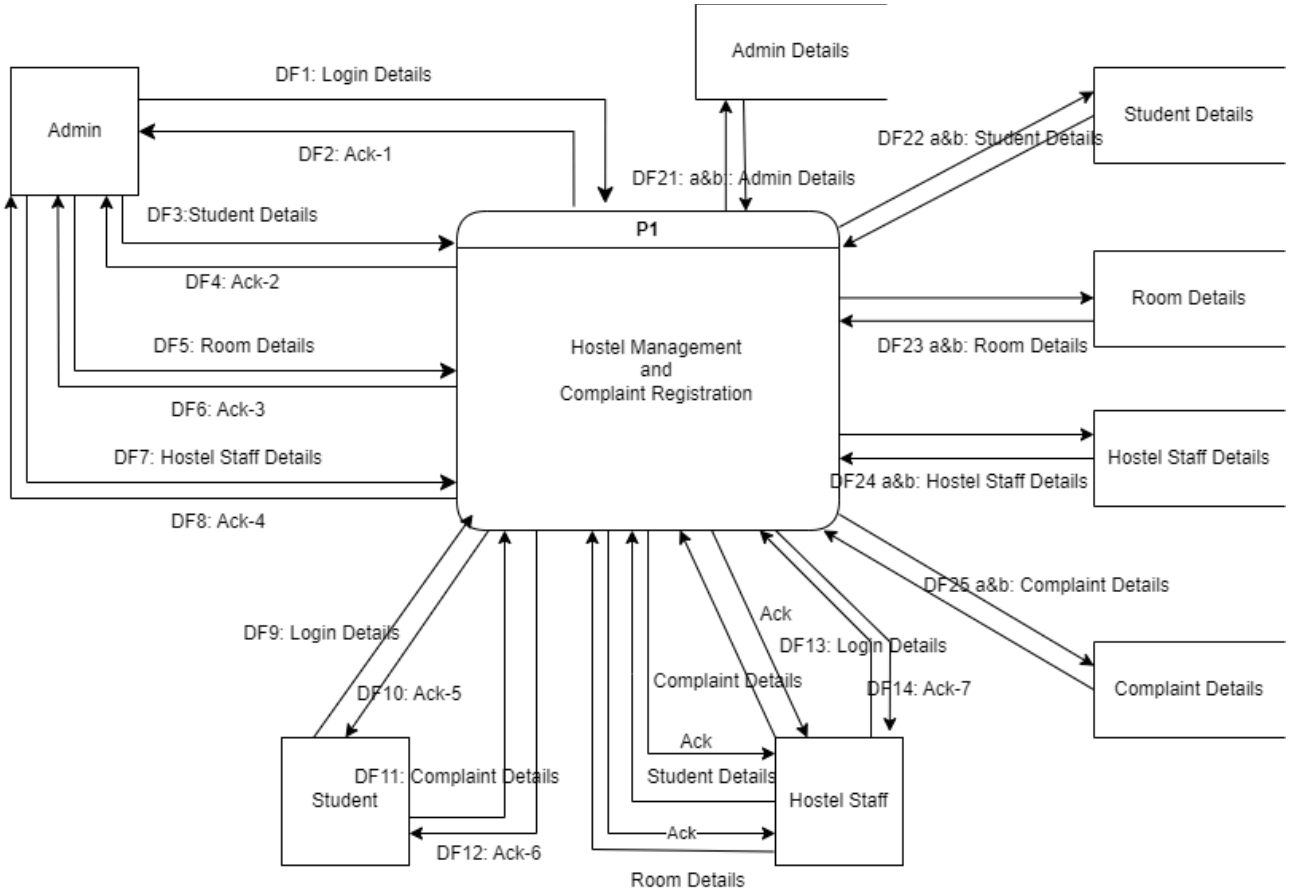

Class Diagram:

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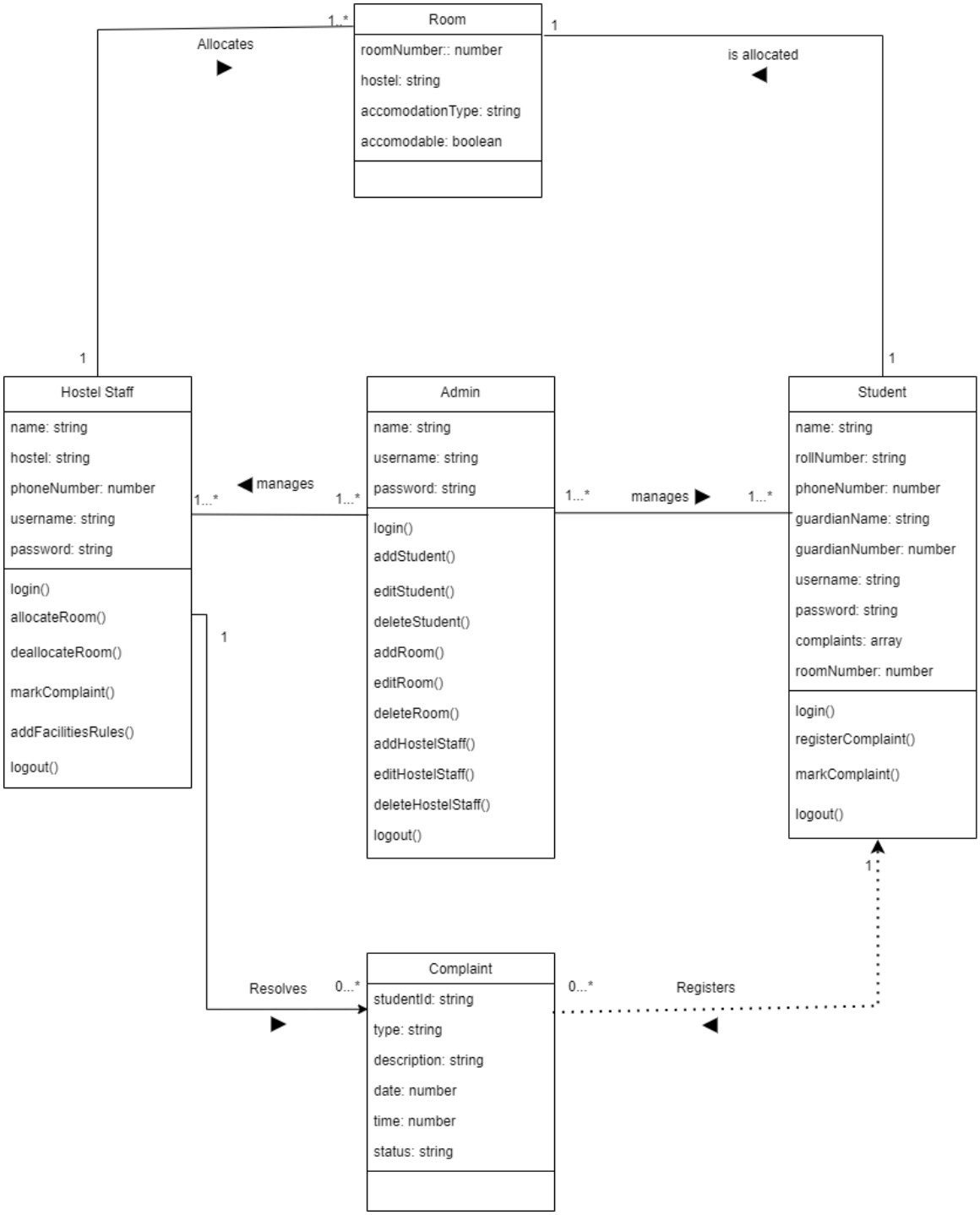

E-R Diagram:

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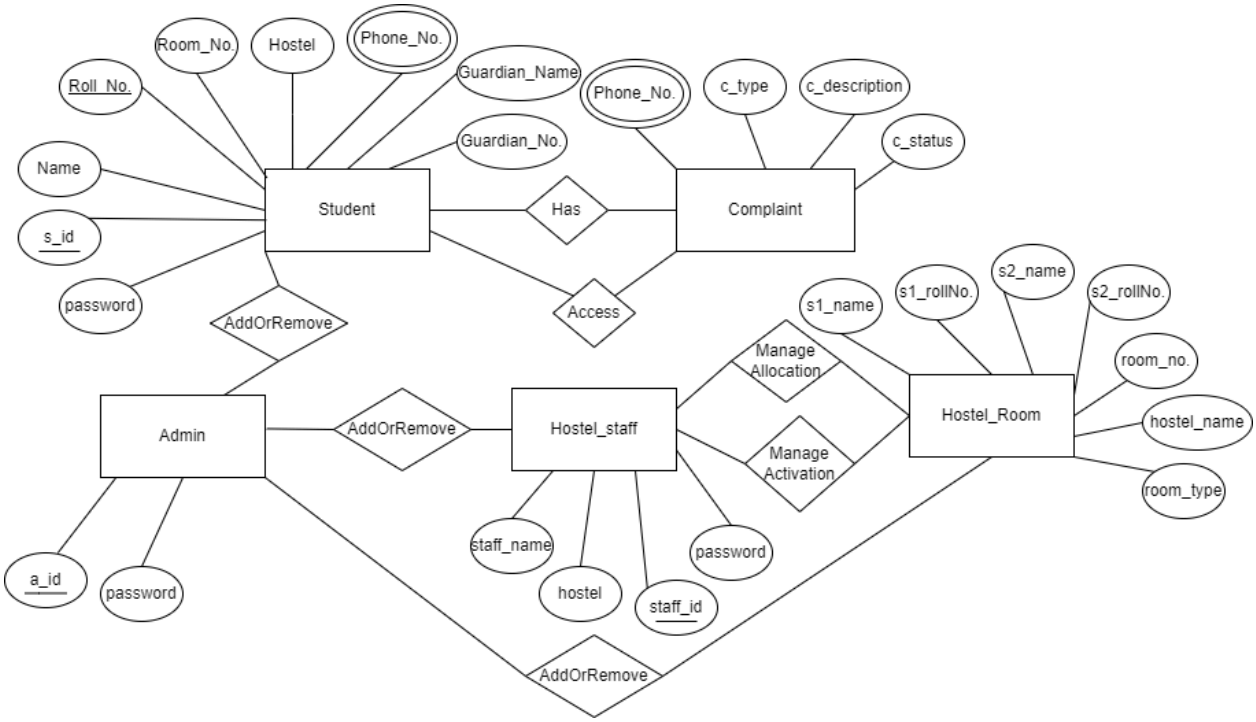

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